

University Knowledge Assistant (RAG-Powered Chatbot)

Problem Statement

Students often struggle to find information about courses, regulations, timetables, examinations, placements, and university policies. The goal of this project is to build a Retrieval-Augmented Generation (RAG) chatbot that can retrieve information from university documents and provide accurate answers.

1. Introduction

The University Knowledge Assistant is an AI-powered chatbot designed to help students access university-related information quickly and efficiently. Instead of manually searching through PDFs and websites, students can ask questions in natural language and receive precise answers.

2. Objectives

- Provide instant answers to student queries.
- Reduce administrative workload.
- Improve access to academic and placement information.
- Support natural language interactions.
- Deliver accurate responses using university documents.

3. System Architecture

Data Sources → Document Processing → Vector Database → Retriever → LLM → User Interface

4. Technology Stack

- Python
- LangChain
- FAISS / ChromaDB
- OpenAI, Gemini, or Llama Models
- FastAPI / Flask Backend
- Streamlit or React Frontend

5. Methodology

Step 1: Collect university documents such as regulations, timetables, placement records, and course handbooks.

Step 2: Extract text from PDFs and documents.

Step 3: Split documents into chunks.

Step 4: Generate embeddings and store them in a vector database.

Step 5: Retrieve relevant chunks based on user queries.

Step 6: Use an LLM to generate contextual responses.

6. Features

- Course Information Retrieval
- Timetable Assistance
- Placement Query Support
- Regulation and Policy Search
- Multi-document Question Answering
- Context-Aware Conversations

7. Expected Outcomes

Students can obtain reliable information instantly, improving productivity and reducing confusion regarding university procedures.

8. Future Enhancements

- Voice-based interaction
- Multilingual support
- Mobile application integration
- Personalized student dashboards
- Real-time university notification integration

9. Conclusion

The University Knowledge Assistant demonstrates how RAG technology can improve access to institutional knowledge. By combining document retrieval with large language models, the system provides accurate, scalable, and user-friendly support for students.